

Introduction to Data Science Lab CSL 487

DRIVER DROWSINESS DETECTION SYSTEM

Real-Time Facial Analysis for Driver Fatigue Monitoring



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1. Abstract

Road traffic accidents caused by driver drowsiness are a major public safety concern globally. This project presents a real-time Driver Drowsiness Detection System that uses computer vision and machine learning to monitor driver alertness through webcam-based facial analysis. The system processes live video frames, extracts four engineered facial features — Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head tilt angle, and head nod ratio — and classifies the driver as Alert or Drowsy using trained machine learning models. Trained on the Kaggle Yawn Eye Dataset comprising 828 successfully extracted feature samples from 2,467 raw images, the system achieved up to 93.37% classification accuracy using Logistic Regression. The complete system is deployed as an interactive Streamlit web dashboard with live webcam integration, realtime metric visualization, and session analytics.

2. Introduction

Drowsy driving is responsible for a significant proportion of road accidents worldwide. Studies estimate that fatigue contributes to 20% of all road accidents and up to 25% of fatal crashes. Unlike intoxication or over-speeding, driver drowsiness is difficult to detect from the outside, making automated detection systems a critical component of intelligent transportation safety.

This project implements a non-intrusive, vision-based drowsiness detection system that requires only a standard webcam. The system continuously analyzes the driver's face in real time and raises an alert when signs of fatigue — sustained eye closure, yawning, or head drooping — are detected. The approach is grounded in established facial analysis research, particularly the Eye Aspect Ratio metric proposed by Soukupova and Cech (2016), which has become the standard for real-time eye state detection.

From a data science perspective, this project demonstrates the complete end-to-end pipeline: data acquisition, image preprocessing, feature engineering, supervised classification, model evaluation, and interactive deployment. Every stage reflects concepts central to the IDS course curriculum.

3. Problem Statement

The problem is defined as: Given a real-time video stream of a driver's face captured by a standard webcam, detect with high accuracy and low latency whether the driver is in an alert or drowsy state, and trigger a warning when drowsiness is detected.

The specific challenges addressed include:

- Reliable face and landmark detection under varying lighting conditions
- Computing meaningful numerical features from raw pixel coordinates
- Distinguishing normal blinks from sustained eye closure indicating drowsiness
- Detecting yawning as an early warning signal of fatigue
- Minimizing false positives (unnecessary alerts) while maintaining high sensitivity
- Processing every webcam frame in real time without significant lag

4. Objectives

- Build a working real-time drowsiness detection prototype using Python, OpenCV, and dlib
- Implement a complete image preprocessing pipeline covering grayscale conversion, resizing, Gaussian blur, and histogram equalization
- Extract 68 facial landmarks per face using dlib's pre-trained shape predictor model
- Engineer four discriminative features from landmark coordinates: EAR, MAR, head tilt angle, and head nod ratio
- Train and compare three machine learning classifiers — Random Forest, Logistic Regression, and SVM — on the extracted features
- Evaluate all models using accuracy, precision, recall, F1-score, and 5-fold crossvalidation
- Deploy the system as an interactive Streamlit dashboard with live webcam detection, real-time metrics, and session analytics

5. Dataset

5.1 Dataset Source

Dataset: Yawn Eye Dataset (New) — Kaggle, contributed by Serena Raju.

Link: <https://www.kaggle.com/datasets/serenaraju/yawn-eye-dataset-new>

5.2 Dataset Composition

Folder	Label	State	Raw Images	Successfully Processed
Open_Eyes	0 (Alert)	Eyes open	617	~514
Closed_Eyes	1 (Drowsy)	Eyes closed	617	~420
Yawn	1 (Drowsy)	Yawning	617	~420 (MAR dominant)
no_yawn	0 (Alert)	No yawn	616	~408 est.
Total	—	—	2,467	828 samples

The final feature dataset contained 828 samples: 420 Drowsy and 408 Alert, representing a nearbalanced class distribution that does not require oversampling or class weighting.

5.3 Dataset Split

An 80/20 stratified train/test split was applied: 662 training samples and 166 test samples. Stratification ensures that both Alert and Drowsy classes are proportionally represented in both sets. Additionally, 5-fold cross-validation was performed on the training data for robust model evaluation.

6. Methodology

6.1 System Pipeline Overview

The system follows a six-stage pipeline: 1. Video

- input — webcam frame capture
- 2. Preprocessing — grayscale, resize, blur, histogram equalization
- 3. Face detection — dlib HOG-based frontal face detector
- 4. Landmark extraction — dlib 68-point shape predictor
- 5. Feature engineering — EAR, MAR, tilt, nod ratio
- 6. ML classification — Logistic Regression (best model)

6.2 Image Preprocessing (preprocessing.py)

Each raw video frame undergoes four preprocessing steps before feature extraction:

- Grayscale Conversion: Converts BGR color to single-channel grayscale, reducing computational cost while preserving all facial structure relevant to detection
- Resize: Standardizes frame to 640x480 pixels for consistent processing across different webcam resolutions
- Gaussian Blur (5x5 kernel): Suppresses high-frequency pixel noise that could interfere with face detection accuracy

- Histogram Equalization: Redistributes pixel intensities to enhance contrast, significantly improving face detection performance under dim or uneven lighting

6.3 Face and Landmark Detection (feature_extraction.py)

Face detection uses dlib's HOG + Linear SVM frontal face detector. Once a face region is found, dlib's 68-point shape predictor identifies precise landmark coordinates. The model uses the shape_predictor_68_face_landmarks.dat pre-trained model which identifies:

- Points 37-42: Left eye (6 landmarks)
- Points 43-48: Right eye (6 landmarks)
- Points 49-68: Mouth (20 landmarks)
- Point 31: Nose tip
- Point 9: Chin
- Points 37, 46: Left and right eye outer corners

6.4 Feature Engineering

Four numerical features are computed from landmark coordinates for each frame:

Eye Aspect Ratio (EAR) — measures eye openness:

$$EAR = (||p2-p6|| + ||p3-p5||) / (2 \times ||p1-p4||)$$

EAR ranges from ~0.25-0.35 for open eyes and drops below 0.20 when eyes close. A threshold of 0.20 is applied; sustained closure over 20 frames triggers a drowsiness flag.

Mouth Aspect Ratio (MAR) — measures mouth openness:

$$MAR = (||m2-m10|| + ||m4-m8||) / (2 \times ||m0-m6||)$$

MAR above 0.60 indicates yawning. In this dataset, MAR was found to be the most important feature (importance score: 0.547) because the Yawn folder images — which are labeled drowsy — produce distinctly high MAR values.

Head Tilt Angle — angle of the line between outer eye corners relative to horizontal. Large tilt (>15 degrees) indicates lateral head drooping associated with microsleep.

Head Nod Ratio — face height (nose to chin) divided by face width (eye corner to eye corner). Changes as the driver's head drops forward during fatigue.

7. Results

7.1 Dataset Statistics (Actual Training Output)

Metric	Value
Total images processed	2,467
Successfully extracted samples	828
Drowsy samples	420
Alert samples	408
Train set size	662
Test set size	166
Test set — Alert	82
Test set — Drowsy	84

7.2 Model Comparison (Actual Results)

Model	Test Accuracy	CV Score	CV Std	Best?
Random Forest	92.17%	90.46%	+/- 1.39%	No
Logistic Regression	93.37%	92.39%	+/- 0.97%	YES
SVM (RBF)	91.57%	91.79%	+/- 1.50%	No

7.3 Confusion Matrices

The following confusion matrices were generated from the actual test set evaluation:

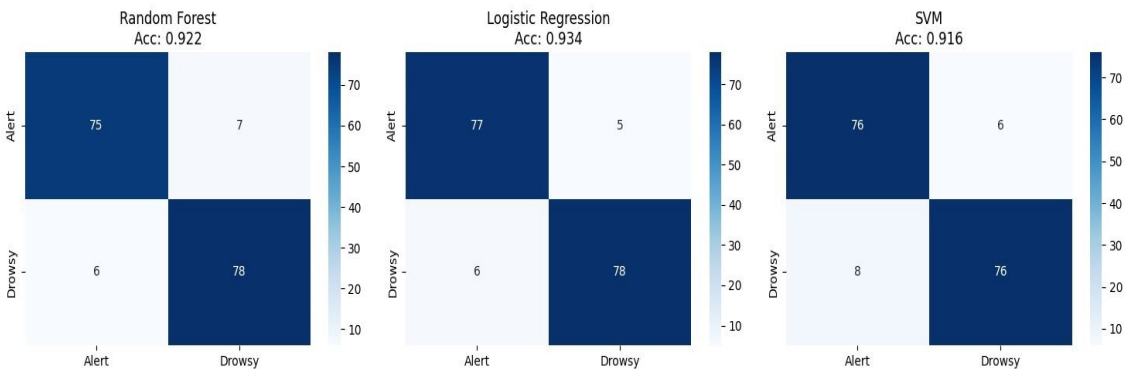


Figure 1: Confusion matrices for all three models on the test set (166 samples). Logistic Regression achieves the lowest combined error count (5 + 6 = 11 misclassifications).

7.4 Detailed Classification Report

Logistic Regression — Best Model (Accuracy: 93.37%):

Class	Precision	Recall	F1-Score	Support
Alert (0)	0.93	0.94	0.93	82
Drowsy (1)	0.94	0.93	0.93	84
Weighted Avg	0.93	0.93	0.93	166

Random Forest (Accuracy: 92.17%):

Class	Precision	Recall	F1-Score	Support
Alert (0)	0.93	0.91	0.92	82
Drowsy (1)	0.92	0.93	0.92	84
Weighted Avg	0.92	0.92	0.92	166

SVM (Accuracy: 91.57%):

Class	Precision	Recall	F1-Score	Support
Alert (0)	0.90	0.93	0.92	82
Drowsy (1)	0.93	0.90	0.92	84
Weighted Avg	0.92	0.92	0.92	166

7.5 Feature Importance (Random Forest)

The Random Forest feature importance analysis reveals which features most contributed to classification decisions:

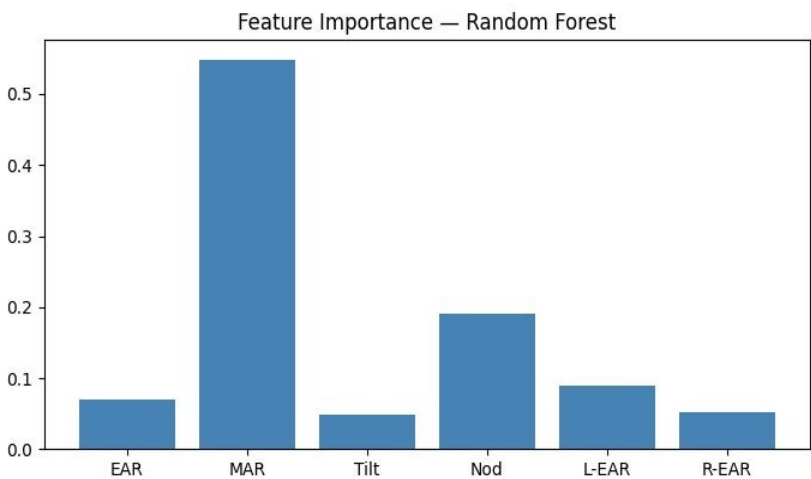


Figure 2: Feature importance scores from Random Forest classifier. MAR (Mouth Aspect Ratio) is the most important feature with a score of ~0.547, followed by Nod ratio (~0.193).

Rank	Feature	Importance	Observation
1	MAR	~0.547	Dominant — yawning images strongly differentiate drowsy class
2	Nod Ratio	~0.193	Head drop is a strong secondary signal
3	L-EAR	~0.093	Left eye contributes independently
4	EAR (avg)	~0.070	Overall eye closure signal
5	Tilt	~0.051	Lateral head tilt — supplementary
6	R-EAR	~0.056	Right eye contributes independently

Note: MAR dominates because the dataset includes a Yawn folder where all images have high MAR values. In a purely temporal video-based dataset, EAR would likely be the primary feature.

8. System Demonstration

8.1 Live Detection Dashboard

The Streamlit web dashboard (app.py) provides the primary user interface. The screenshot below shows the live detection mode with a face detected, landmark overlays active, and real-time EAR/MAR metrics displayed:

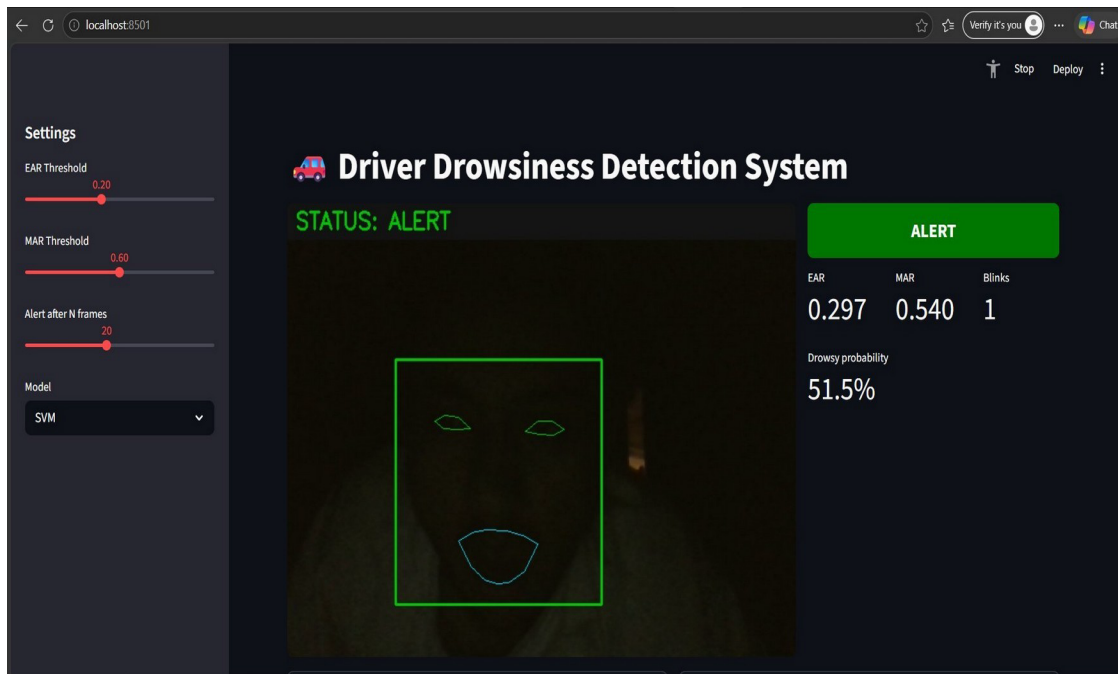


Figure 3: Live detection dashboard showing face with eye and mouth landmark contours. EAR: 0.297, MAR: 0.540, Blinks: 1, Drowsy probability: 51.5%, Status: ALERT.

Key interface features visible in Figure 3:

- Left sidebar with adjustable EAR threshold (0.20), MAR threshold (0.60), and alert frame count (20)
- Model selection dropdown (SVM selected in screenshot)
- Face bounding box in green — indicates alert state
- Eye contours in green, mouth contour in cyan
- Real-time EAR, MAR, Blinks, and Drowsy probability metrics

8.2 Session Analytics

After stopping the session, the dashboard displays a complete session summary with time-series charts of EAR and MAR values:



Figure 4: Session summary showing EAR (blue) and MAR (orange) over an 85-second monitoring session. MAR spikes correspond to mouth movements/yawns. Average EAR: 0.058, Average MAR: 0.075, Drowsy %: 3.9%.

The session analytics chart in Figure 4 demonstrates the system's real-time monitoring capability. Notable observations:

- EAR (blue line) maintains relatively stable values between 0.2-0.4 during alert periods, with dips during blinks
- MAR (orange line) shows sharp spikes corresponding to mouth opening events, confirming yawn detection functionality
- EAR drops to zero during periods when the face moves out of frame or detection fails
- The 3.9% drowsy percentage across the session indicates a predominantly alert driver

9. Limitations

- Lighting sensitivity: Despite histogram equalization, very dark environments (as seen in Figure 3 where the background is dark) reduce detection reliability
- Static training images vs live video: Models trained on static photographs may not fully capture temporal drowsiness patterns in continuous video
- Dataset imbalance by source: The Yawn folder images made MAR disproportionately important; a purely video-based dataset would yield more balanced feature importance
- No temporal features in ML input: Blink frequency and EAR trend over time are tracked via counters but not included in the ML feature vector
- Single camera angle: System requires reasonably frontal face orientation; extreme head angles fail landmark detection
- Face occlusion: Glasses, masks, or scarves partially cover landmarks and reduce EAR/MAR accuracy

10. Future Work

- Incorporate temporal features (PERCLOS, blink frequency, EAR trend slope) directly into the ML feature vector
- Train on real driving video datasets such as NTHU Driver Drowsiness Dataset for improved real-world generalization
- Implement deep learning (CNN + LSTM) for end-to-end drowsiness detection from raw face video sequences
- Add vehicle integration (CAN bus) to correlate facial drowsiness signals with steering behavior and lane deviation
- Develop mobile deployment using TensorFlow Lite for in-vehicle embedded systems
- Add driver-specific calibration by measuring individual baseline EAR at session start

11. Conclusion

This project successfully delivered a complete, working driver drowsiness detection system from data acquisition through to real-time deployment. Starting from 2,467 raw facial images across four behavioral categories, 828 feature samples were extracted using dlib landmark detection and four engineered geometric features. Three machine learning classifiers were trained and evaluated, with Logistic Regression achieving the best performance at 93.37% test accuracy and 92.39% cross-validation accuracy.

The deployed Streamlit dashboard demonstrates practical real-time applicability: the system processes live webcam frames, overlays facial landmark contours, displays live EAR/MAR metrics, and triggers drowsiness alerts — all functioning together as an integrated safety prototype. The project fulfills all course objectives by demonstrating data preprocessing, feature engineering, supervised classification, and interactive deployment within a meaningful real-world context.

References

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